UNIVERSITI TEKNOLOGI MALAYSIA

SEMESTER 1

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SECJ3553-02 ARTIFICIAL INTELLIGENCE (AI)

**Assignment 1**

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## Introduction

In this assignment, we focus on the Ideate phase of Design Thinking by planning how our AI system will work. We describe the Knowledge Representation (KR) needed for the Smart Waste Management System and explain how the AI uses this knowledge to make decisions. Using simple logical rules and First-Order Logic (FOL), we show how the system can predict bin levels, give alerts, and plan better waste collection routes. This helps support our goal of creating a cleaner and more efficient smart city.

## Knowledge Representation

### 1. Knowledge Representation #1 Bin Fill-Level Knowledge

This KR stores knowledge about the waste bin's real-time status (fill level, weight, disposal frequency). It allows the AI system to recognize when a bin is almost full and requires attention.

FOL Representation

Let :

- Bin (x) = x is a waste bin
- FillLevel(x,y) = bin x has fill level y (percentage)
- NearFull(x) - bin x is near full
- Threshold(t) = threshold percentage (70%)

Natural Language : IF x is a bin AND the fill level of x is y AND the threshold is t AND y is greater than or equal to t THEN the bin x is considered near full.

$$\forall x \forall y \forall t[\mathrm{Bin}(x) \land \mathrm{FillLevel}(x,y) \land \mathrm{Threshold}(t) \land y \geq t \rightarrow \mathrm{NearFull}(x)]$$

If a bin reaches or exceeds the threshold (70%), the system marks it as ''near full.'

This KR is used by the predictive model to detect which bins need immediate collection. It helps the system maintain cleanliness and prevent overflow.

### 2. Knowledge Representation #2 Waste Accumulation Prediction Knowledge

This KR represents patterns of waste accumulation based on time, zone population, events, and historical data. The AI model uses this knowledge to predict when the bin will be full.

FOL Representation

Let :

- HighUsageZone(z) = zone z produces waste quickly
- BinInZone(x,z) = bin x is located in zone z
- PredictFull(x,t) = bin x will be full at time t

Natural Language : IF x is a bin AND the bin is located in zone z AND zone z is a high waste usage zone THEN the system predicts that bin x will become full at time t.

$$\forall x \forall z \forall t[Bin(x) \wedge BinInZone(x,z) \wedge HighUsageZone(z) \rightarrow PredictFull(x,t)]$$

Bins in busy zones fill faster, so the system predicts earlier full times.

This KR allows the AI to forecast fill-level trends and schedule collection before overflow, helping achieve the goal of cleanliness and efficiency.

3. **Knowledge Representation #3 Collection Priority Knowledge**

This KR tells the system which bins should be collected first based on urgency.

FOL Representation

Let:

- NearFull(x) = bin x in near full
- Urgent(x) = bin x needs immediate collection
- EventDay(x) = today is a special event (more wasted generated)

Natural Language :

1. IF bin x is near full THEN bin x becomes urgent :

$$\forall x \, [NearFull(x) \rightarrow Urgent(x)]$$

2. IF bin x has medium fill level AND today is an event day THEN bin x is urgent :

   Let MediumFill(x) = 50-69%

$$\forall x \forall d[MediumFill(x) \wedge EventDay(d) \rightarrow Urgent(x)]$$

During events, waste increases so even medium-level bins become a high priority.

This KR allows the autonomous system to decide priority and reduce complaints by ensuring timely collection.

4. **Knowledge Representation #4 Route Optimization Knowledge**

This KR tells the system as AI needs knowledge of the distances, locations and travel time to generate the best route.

FOL Representation

Let :

- Bin(x)
- Distance(x,y,z) = distance between locations x and y is d
- Route(a,b,r) = route r connects a to b
- Shortest(r) = r is the shortest route

Natural Language : IF r is a route from location a to location b AND route r is the shortest route THEN r is the optimal route.

$$\forall a \forall b \forall r[\text{Route}(a,b,r) \wedge \text{Shortest}(r) \rightarrow \text{OptimalRoute}(r)]$$

The AI must always select the shortest route for efficiency.

This KR helps reduce fuel usage, travel time and operational cost which directly supporting the project goal

5. **Knowledge Presentation #5 Vehicle Assignment Knowledge**

This KR tells the system about availability of vehicles and assigning the nearest one to an urgent bin.

FOL Representation

Let:

- Vehicle(v)
- Available(v)
- Near(v, x)

- Assign(v, x)

Natural Language: IF v is a vehicle AND v is available AND bin x is urgent AND vehicle v is near bin x THEN assign vehicle v to service bin x.

$$\forall v \forall x[Vehicle(v) \land Available(v) \land Urgent(x) \land Near(v, x) \rightarrow Assign(v,x)]$$

The nearest available truck must respond to urgent bins.

This KR reduces response time and ensures efficiency in waste handling.

**Supporting KR to The Project Goal**

Table 1 shows the Knowledge Representations (KRs) that support the project goal by helping the AI optimize waste collection schedule, reduce operational cost and improve sustainability. KRs #1, #2, #3 ensure the system to detect full bins, predict waste levels and prioritize urgent bins to keep the city clean and optimize the collection schedule. Meanwhile, KRs #4 and #5 allow the AI to plan efficient routes and assign the right vehicles, which lowers operational costs. Altogether, KRs #1 until #5 work together to reduce overflow, minimize fuel usage and support long-term sustainability for city waste management.

Table 1 Supporting KR to the goals

| **Goal** | **Supporting KR** |
|---|---|
| Optimize waste collection schedule | KR #1, #2, #3 |
| Reduce operational cost | KR #4, #5 |
| Improve sustainability | ALL |

**Conclusion**

In this assignment, we explored how the Ideate phase of Design Thinking supports the development of a Smart Waste Management System through effective Knowledge Representation (KR). By defining key KRs using natural language rules and First-Order Logic (FOL), we demonstrated how the AI system can recognize bin status, predict waste accumulation, prioritize collection, optimize routes and assign vehicles efficiently. These representations work together to improve cleanliness, reduce operational costs and promote sustainability in smart city waste management. Overall, the KR framework strengthens the system's decision making ability and ensures a more responsive and intelligent waste management process.